

UEC634 : CLOUD COMPUTING TECHNOLOGY

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2	0	2	3.0

Course Objective:

To appreciate the benefits of Cloud computing and apply Cloud paradigms for evolving businesses. To familiarize with cloud architectural models and resource allocation strategies. The student should comprehensively be exposed to Cloud based services.

Introduction: Basics of the emerging Cloud computing paradigm, Cloud computing history and evolution, Cloud enabling technologies, practical applications of Cloud computing for various industries, the economics and benefits of Cloud computing.

Cloud Computing Architecture: Cloud Architecture model, Types of Clouds: Public private & Hybrid Clouds, Resource management and scheduling, QoS (Quality of Service) and Resource Allocation, Clustering.

Cloud Computing delivery Models: Cloud based services: IaaS, PaaS and SaaS Infrastructure as a Service (IaaS): Introduction to IaaS, Resource Virtualization i.e. Server, Storage and Network virtualization Platform as a Service (PaaS): Introduction to PaaS, Cloud platform & Management of Computation and Storage, Azure, Hadoop, and Google App. Software as a Service (SaaS): Introduction to SaaS, Cloud Services, Web services, Web 2.0, Web OS Case studies related to IaaS, PaaS and SaaS.

Data Processing in Cloud: Introduction to Map Reduce for Simplified data processing on Large clusters, Design of data applications based on Map Reduce in Apache Hadoop

Advanced Technologies: Advanced web technologies (AJAX and Mashup), distributed computing models and technologies (Hadoop and MapReduce), Introduction to Open Source Clouds like Virtual Computing Lab (Apache VCL), Eucalyptus

Cloud Issues and Challenges: Cloud computing issues and challenges like Cloud provider Lock-in, Security etc.

Laboratory Work :

Cloud Service creation, deploying, resource sharing, and security using AWS Academy resources

Course Learning Objectives (CLO)

The students will be able to:

1. Identify Cloud architectures and Cloud based services.
2. Analyze economic impact of implementing cloud computing technologies
3. Applying data processing and clustering algorithms in cloud
4. Evaluate security issues in the Cloud environment.
5. Implement cloud based services

Text Books

1. Rajkumar Buyya, James Broberg and Goscinski Author Name, Cloud Computing Principles and Paradigms, John Wiley and Sons 2012, Second Edition.
2. Gerard Blokdijs, Ivanka Menken, The Complete Cornerstone Guide to Cloud Computing Best Practices, Emereo Pvt Ltd, 2009, Second Edition.

Reference Books

1. Anthony Velte, Toby Velte and Robert Elsenpeter, Cloud Computing: A practical Approach Tata McGrawHill, 2010, Second Edition
2. Judith Hurwitz, Robin Bllor, Marcia Kaufmann, Fern Halper, Cloud cOmputing for Dummies, 2009, Third Edition